Reading copy: notes 03 and 04 of `lecc-small`

Prepared for Sid, 9 October 2026

This is a typeset copy of notes `03` and `04` in `lecc-small/`, for reading before the 5:00 PM meeting. The Markdown notes are the source of truth. While reading, mark any step you cannot justify in one line.

1 Theorem R5: for regular graphs, the kernel method works only on bipartite graphs

Theorem 1. *Let G be a simple k-regular graph with $k \geq 1$. Then $L(G)$ has a kernel-perfect orientation with maximum out-degree at most $k-1$ if and only if G is bipartite.*

"If" is Galvin's theorem. "Only if" is the direction we prove. Through Lemma 1 of Chapter 33 of *Proofs from THE BOOK*, it says:

Corollary 2. *For a regular graph G that is not bipartite, the kernel method (Lemma 1 with lists of size $\Delta(G)$) cannot prove $\mathrm{ch}'(G) = \Delta(G)$, whatever the true value of $\mathrm{ch}'(G)$ is.*

Special cases it contains. The triangle theorem (note 11), the even complete graph theorem (note 12), the case $C_5 \times K_2$, and the failure of the kernel method on all 52 of Schauz's open zero-sum graphs.

Novelty. It is not in Borodin, Kostochka and Woodall (1998), who characterise kernel-perfect orientations of line graphs, nor in McDonald (2015, arXiv:1508.01820). McDonald proves the K_n case only for her restricted "Galvin orientations", and shows the Petersen graph *does* have one with $4 = \Delta + 1$ colours. So the barrier is specific to the tight budget $k = \Delta$ on regular graphs. Status: **not found in the literature; not yet claimed as new.**

Proof

Throughout, D is a kernel-perfect orientation of $L(G)$ with $d^+(e) \leq k-1$ for every edge e of G.

Step (1: ranks)**.** By the star lemma (note 09), for each vertex u the k edges at u induce a transitive tournament in D. Let $r_u(v)$ be the number of edges at u that uv points to: its *rank* at u, with 0 the best. At each u the ranks are exactly $0, 1, \ldots, k-1$. Since G is simple, two edges share at most one vertex, so every arc of D lies in exactly one star, and

$$d^+(uv) = r_u(v) + r_v(u).$$

Step (2: tightness)**.** Summing over all edges,

$$\sum_{uv \in E} d^+(uv) = \sum_{u \in V} \big(0 + 1 + \cdots + (k-1)\big) = |V| \, \frac{k(k-1)}{2} = |E| \, (k-1).$$

Each term is at most $k-1$, so every term equals $k-1$:

$$r_u(v) + r_v(u) = k - 1 \qquad \text{for every edge } uv.$$

Step (3: no odd ring)**.** An *odd ring* is an odd cycle $a_1a_2\dots a_m$ of G ($m \geq 3$ odd) in which every a_i ranks a_{i+1} above a_{i-1}, indices mod m. Suppose one exists. Let S be its edge set and $e_i = a_ia_{i+1}$.

- In $L(G)$, within S the edge e_i is adjacent only to e_{i-1} and e_{i+1}, because the cycle is simple. (For $m = 3$ all three are adjacent; same statement.)
- At a_i, the edge e_i is ranked above e_{i-1}, so $e_{i-1} \to e_i$.
- Hence $D[S]$ is a directed cycle of odd length, which has no kernel. This contradicts kernel-perfectness.

So G has no odd ring.

Step (4: an orientation of G)**.** Put $y(u, v) = r_u(v) - \frac{k-1}{2}$. By Step 2, $y(v, u) = -y(u, v)$. Orient uv as $u \to v$ in a digraph O when $y(u, v) < 0$, that is, when u ranks v in its better half. Let M_0 be the set of edges with $y = 0$; they stay unoriented.

- If k is even, y is never 0, so $M_0 = \emptyset$.
- If k is odd, $y(u, v) = 0$ exactly when $r_u(v) = \frac{k-1}{2}$, and by Step 2 then also $r_v(u) = \frac{k-1}{2}$. Each vertex has exactly one edge of middle rank, so M_0 is a perfect matching.

At every vertex the out-arcs of O are the edges of rank below $\frac{k-1}{2}$ and the in-arcs those above it, $\lfloor k/2 \rfloor$ of each. So O is a balanced (Eulerian) orientation of $G - M_0$.

Step (5: admissible cycles are rings)**.** Traverse a cycle $a_1 \to a_2 \to \cdots \to a_m \to a_1$. Then a_i ranks a_{i+1} above a_{i-1} exactly when

$$\begin{aligned} r_{a_i}(a_{i+1}) < r_{a_i}(a_{i-1}) &\iff y(a_i, a_{i+1}) < y(a_i, a_{i-1}) = -y(a_{i-1}, a_i) \\ &\iff y(a_{i-1}, a_i) + y(a_i, a_{i+1}) < 0. \end{aligned}$$

Call the cycle *admissible* if every edge is either an arc of O traversed forwards ($y < 0$) or an edge of M_0 ($y = 0$). Two consecutive edges share a vertex, so they are not both in the matching M_0. Hence in an admissible cycle every consecutive sum is negative, and **every admissible odd cycle is an odd ring**. By Step 3, G has no admissible odd cycle.

Step (6: strong connectivity)**.** In a balanced digraph every arc lies on a directed cycle. Hence each connected component H of $G - M_0$, oriented by O, is strongly connected.

Step (7: each component is bipartite)**.** Suppose H is not bipartite.

- *If there is no odd directed closed walk, return paths are odd.* Suppose H has no directed closed walk of odd length. For each arc $x \to y$ pick a directed path P_{yx} from y to x (Step 6). Then $x \to y$ followed by P_{yx} is a closed walk, so $|P_{yx}|$ is odd.
- *Build an odd closed walk.* Take an odd cycle Z of H and walk around it. Keep each arc traversed forwards, and replace each arc $x \to y$ traversed backwards by P_{yx}. The result is a directed closed walk of length $\equiv |Z| \equiv 1 \pmod 2$, a contradiction. So H has an odd directed closed walk.
- *Extract a cycle.* A directed closed walk splits into directed cycles whose lengths add up, so one of them is odd. A directed odd cycle of O is admissible, contradicting Step 5.

So H is bipartite.

Step (8: k even)**.** $M_0 = \emptyset$, so G is the union of its components, each bipartite by Step 7.

Step (9: k odd)**.** Fix a 2-colouring col of every component of $G - M_0$. Form the multigraph Γ whose nodes are the components and whose edges are the edges of M_0, loops allowed. Give $f = vu \in M_0$ the label $\lambda(f) = 1$ if $\mathrm{col}(v) = \mathrm{col}(u)$, and 0 otherwise. Flipping the colouring of a component changes λ on its non-loop edges, so

$$G \text{ is bipartite} \iff \text{every cycle of } \Gamma \text{ has even label sum.}$$

Suppose some cycle of Γ has odd label sum; take one with the fewest edges.

- *It visits no component twice*, or it would split into two shorter closed walks, one with odd sum. Write it as $H_1, \dots, H_t$, distinct, with $f_i = v_iu_{i+1}$ joining H_i to H_{i+1} (indices mod t; $t = 1$ is a loop).

- *Entry and exit differ:* in H_i we enter at u_i and leave at v_i, and $u_i \neq v_i$ because M_0 is a matching.
- *A cycle in G:* by Step 6 choose a directed O-path P_i from u_i to v_i inside H_i. Then $Z = P_1 f_1 P_2 f_2 \dots P_t f_t$ is a simple cycle of G, since the components are distinct, and it is admissible.
- *It is odd:* H_i is bipartite, so $|P_i| \equiv \mathrm{col}(u_i) + \mathrm{col}(v_i) \pmod 2$. Regrouping cyclically,

$$|Z| = \sum_i |P_i| + t \equiv \sum_i \big(\mathrm{col}(v_i) + \mathrm{col}(u_{i+1}) + 1\big) \equiv \sum_i \lambda(f_i) \equiv 1 \pmod 2.$$

So Z is an admissible odd cycle, contradicting Step 5. Therefore G is bipartite. □

Remarks

- **Not needed:** Tan or Chung (the hard direction of the odd-ring lemma, note 02), or the class-1 hypothesis. Only the star lemma, counting, and "a directed odd cycle has no kernel".
- **Simple graphs only.** Step 1 uses simplicity. The multigraph case is open here.
- **Sharpness.** Regularity forces tightness; for non-regular graphs McDonald's "bipartite plus an edge" shows the kernel method can succeed off the bipartite world. The budget must be $k = \Delta$: with $\Delta + 1$ the Petersen graph succeeds (McDonald, Theorem 4).
- **Machine confirmation:** `kernel_sat.py` on 1923 connected regular graphs (6, 8 and 10 vertices, cubic on 12, triangle-free cubic on 14, 4-regular on 12). SAT on exactly the 29 bipartite ones.

2 Computer proofs of the LECC for open zero-sum graphs

Claim. For each graph in the table at the end, $\mathrm{ch}'(G) = \chi'(G) = k$, that is, $L(G)$ is k-choosable. For each of them the uniform Alon–Tarsi coefficient vanishes, so Schauz's method fails, and by Theorem R5 Galvin's kernel method fails.

Setting. $H = L(G)$, n vertices, lists of size k. A list assignment is *bad* if H has no proper colouring from it. We show no bad assignment exists.

Lemma 3 (P, precheck)**.** *If for every vertex v the graph $H - v$ has a monomial with nonzero coefficient and all exponents at most $k-1$, then by the Combinatorial Nullstellensatz every proper induced subgraph of H is k-choosable.*

Lemma 4 (C, criticality)**.** *Under (P), let L be bad, v a vertex and $c \in L(v)$. Then c lies in the list of some neighbour of v.*

Proof. Otherwise colour $H - v$ from L (possible by (P)) and give v the colour c. □

Lemma 5 (B, palette)**.** *Under (C), every colour lies in at least 2 lists. There are nk list slots, so at most $nk/2$ colours occur; rename them into $\{0, \dots, nk/2 - 1\}$.*

Lemma 6 (S, symmetry)**.** *Rename colours in order of first appearance along $v_0, v_1, \dots$. Then whenever colour $c \geq 1$ lies in $L(v_i)$, colour $c-1$ lies in some $L(v_j)$ with $j \leq i$. Renaming does not change badness, (C) or (R).*

Lemma 7 (R, reducible colour classes)**.** *Let $A(c)$ be the set of vertices whose list contains c. Suppose $I \subseteq A(c)$ is independent and $H - I$ has a nonzero monomial x^t with $t_w \leq k-2$ for $w \in A(c) \setminus I$ and $t_w \leq k-1$ elsewhere. Then L is colourable.*

Proof. Colour I with c and delete c from the lists of $A(c) \setminus I$. Those lists now have size $k - 1 > t_w$, and the others size $k > t_w$. The Combinatorial Nullstellensatz colours $H - I$ from the reduced lists, none of which contains c. □

The condition depends only on the set $A(c)$. So in a bad L no colour class A satisfies $I \subseteq A \subseteq I \cup Z(t)$, where $Z(t) = \{w : t_w \leq k-2\}$. Each such pair (I, t) is a *witness*.

The loop (CEGAR). A SAT solver searches for an assignment satisfying "lists of size k", (B), (S), (C) and (R), and admitting none of the colourings found so far. If it proposes L, a second solver looks for colourings from L; if there are some, up to 8 are added as exclusions and the loop repeats. If it reports UNSAT, no bad assignment exists, so H is k-choosable.

Certificates (new, 9 October). The search was rerun with one selector literal per witness (`core_check.py`). After UNSAT, the solver's unsat core names the witnesses the refutation actually used. `verify_core.py`, which shares no code with the search, then re-checks:

1. (P): each stored monomial is the out-degree sequence of an orientation of $H-v$ with $EE \neq EO$ (Alon–Tarsi, counted by Eulerian subgraphs, not by expanding the polynomial);
2. (R): the same for every core witness, plus I independent and $Z(t)$ as claimed;
3. every blocked colouring is proper;
4. the CNF rebuilt from (L), (S), (C), the core witnesses and the exclusions is UNSAT under a different solver (Glucose 4; the search used CaDiCaL).

Validation of the method. `tests/test_choosability.py` compares against brute force over every list assignment on 79 random graphs (4 to 6 vertices), and compares `-reduce` with the plain search on 120 more (6 to 9 vertices). Known cases come out right, including non-choosability of L(Petersen) with 3 colours.

Results.

graph6	N	k	core / all witnesses	independent check
`GCY^B_` (X)	8	3	892 / 6972	passed
`I?`amQoM?`	10	3	4636 / 73657	passed
`I?`bfAWF?`	10	3	4473 / 75994	passed
`ICOedO[X?`	10	3	5809 / 73302	passed
`I?bEHow[?`	10	3	7723 / 70276	passed
`GEnbvG`	8	4	2235 / 201177 (lazy)	passed

Still open. A DRAT proof of the final UNSAT checked by `drat-trim` (so the SAT step is certified, not just reproduced by a second solver), and a human reading of (P), (C), (B), (S), (R).

Degree 4 (9 October). Adding all (R) witnesses up front gives 6.4 million clauses for `GEnbvG` and the solver never finished its first call. `lazy_choos.py` keeps the witnesses in memory and, for each proposed L, adds clauses only for witnesses matching its actual colour classes. Soundness is unchanged: every clause added is still a valid witness. Result: `GEnbvG` is 4-edge-choosable after 30 seconds of search (plus 18 minutes to generate the witnesses), and its certificate passes `verify_core.py`.